

Learning Stable Latent Degradation Manifolds from Multivariate Time Series

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Abstract—Multivariate degradation time series are often collected under varying operating regimes, making direct forecasting unstable and difficult to interpret. We propose a workflow for learning stable latent degradation manifolds from such data. The method identifies operating regimes from control variables, normalizes sensor measurements within regimes, denoises the resulting time series, selects degradation-sensitive sensors, and trains a bounded latent autoencoder on the dynamic denoised channels. The learned latent space provides a compact degradation state suitable for stable long-horizon rollout and downstream remaining-useful-life prediction. Experiments show that increasing latent dimension improves reconstruction but can reduce rollout stability, suggesting that low-dimensional bounded manifolds act as prognostic regularizers rather than simple bottlenecks.

Index Terms—degradation modeling, latent manifold, autoencoder, remaining useful life, stable forecasting, multivariate time series

I. INTRODUCTION

Multivariate degradation data are common in engineered systems, industrial monitoring, and prognostics. Such data are challenging because sensor measurements are affected both by latent degradation and by external operating conditions. As a result, direct forecasting in sensor space can be unstable, especially when models are rolled out freely over long horizons.

Most predictive models emphasize short-horizon accuracy, reconstruction quality, or remaining-useful-life estimation. However, prognostic systems also require stable long-horizon behavior. A model that reconstructs historical trajectories accurately may still produce implausible or divergent forecasts when recursively rolled forward.

This paper proposes a stable latent degradation manifold for multivariate time series. Instead of forecasting raw sensors directly, we learn a bounded low-dimensional latent state that captures degradation-relevant structure after removing operating-regime effects and suppressing sensor noise.

The main contributions are:

- A regime-aware workflow for learning latent degradation states from multivariate time series.
- A bounded latent autoencoder objective with monotonicity and smoothness regularization on the primary health coordinate.
- A theoretical analysis showing that bounded latent rollouts produce bounded decoded forecasts, and that smooth latent trajectories reduce local forecast error.

- An experimental evaluation showing that higher-dimensional latent spaces can improve reconstruction while reducing rollout stability, supporting low-dimensionality as a prognostic regularizer.
- Comparisons against autoregressive, recurrent, specialist, and generalized variants.

II. RELATED WORK

A. Health Indicators and Degradation Modeling

Prior work on prognostics often constructs health indicators that are monotonic, trendable, and prognosable. These indicators are then used for fault tracking, degradation forecasting, or remaining-useful-life prediction.

B. Autoencoder-Based Representation Learning

Autoencoders and variational autoencoders have been widely used to learn low-dimensional representations of multivariate sensor data. However, reconstruction-oriented objectives alone do not guarantee stable long-horizon behavior.

C. Latent Dynamics and Sequence Models

Autoregressive models, recurrent neural networks, GRUs, and LSTMs are common baselines for time-series forecasting. These models can achieve strong short-horizon performance, but their free-run stability is not guaranteed.

D. Constrained and Stable Forecasting

This work is related to constrained representation learning and stable latent dynamics. Our focus is not simply on learning a compact representation, but on learning a latent geometry that supports bounded degradation rollout.

III. METHODOLOGY

A. Problem Formulation

Let $\mathbf{x}_{i,t} \in \mathbb{R}^p$ denote the vector of sensor measurements for unit i at time t , and let $\mathbf{u}_{i,t} \in \mathbb{R}^q$ denote the corresponding control or operating variables. The goal is to learn a low-dimensional latent state

$$\mathbf{z}_{i,t} \in \mathbb{R}^K, \quad K \ll p,$$

that captures degradation-relevant dynamics and supports stable long-horizon forecasting.

B. Workflow Overview

The proposed workflow consists of five main stages:

- 1) regime identification from control variables;
- 2) regime normalization of sensor variables;
- 3) denoising of normalized sensor trajectories;
- 4) selection of degradation-sensitive dynamic sensors;
- 5) bounded latent autoencoder training.

C. Regime Identification from Control Variables

Operating regimes are identified using only the control variables $\mathbf{u}_{i,t}$. A clustering function C assigns each observation to a discrete regime:

$$r_{i,t} = C(\mathbf{u}_{i,t}).$$

In our implementation, C is a K-means clustering model fit on the control variables. Sensors are not used for regime identification, preventing degradation-sensitive measurements from influencing the operating-condition partition.

D. Regime Normalization of Sensors

For each regime r and sensor j , we estimate a regime-specific mean $\mu_{r,j}$ and scale $\sigma_{r,j}$. The normalized sensor value is

$$\tilde{x}_{i,t,j} = \frac{x_{i,t,j} - \mu_{r_{i,t},j}}{\sigma_{r_{i,t},j} + \epsilon}.$$

This step removes operating-regime offsets and produces residualized sensor trajectories that are more likely to reflect degradation.

E. Denoising of Normalized Sensor Trajectories

The normalized sensor trajectories are denoised independently for each unit. Let $W(t)$ be a temporal window around time t . We compute

$$\bar{x}_{i,t,j} = \text{median}_{\tau \in W(t)} \tilde{x}_{i,\tau,j}.$$

A rolling median is used because it suppresses local spikes while preserving slowly varying degradation trends.

F. Selection of Degradation-Sensitive Sensors

Not all sensors carry degradation information. For each denoised sensor j , we compute a trend score

$$S_j = \frac{1}{N} \sum_i |\text{corr}(\bar{x}_{i,:}, j, t)|.$$

The dynamic sensor set is defined as

$$\mathcal{S} = \{j : S_j \geq \gamma\},$$

where γ is a fixed trend threshold. The autoencoder is trained only on the selected dynamic sensors:

$$\mathbf{y}_{i,t} = \bar{\mathbf{x}}_{i,t,\mathcal{S}}.$$

G. Bounded Latent Autoencoder

The selected denoised sensor vector $\mathbf{y}_{i,t}$ is mapped to a bounded latent state:

$$\mathbf{z}_{i,t} = f_\theta(\mathbf{y}_{i,t}), \quad \hat{\mathbf{y}}_{i,t} = g_\phi(\mathbf{z}_{i,t}),$$

where f_θ is the encoder and g_ϕ is the decoder. The latent state is bounded by construction:

$$\mathbf{z}_{i,t} \in [0, 1]^K.$$

The training objective is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_m \mathcal{L}_{\text{mono}} + \lambda_s \mathcal{L}_{\text{smooth}}.$$

The reconstruction term is

$$\mathcal{L}_{\text{rec}} = \sum_{i,t} \|\hat{\mathbf{y}}_{i,t} - \mathbf{y}_{i,t}\|_2^2.$$

The monotonicity penalty is applied to the first latent coordinate:

$$\mathcal{L}_{\text{mono}} = \sum_{i,t} \max(0, -\Delta z_{i,t,1}),$$

where

$$\Delta z_{i,t,1} = z_{i,t+1,1} - z_{i,t,1}.$$

The smoothness penalty is

$$\mathcal{L}_{\text{smooth}} = \sum_{i,t} (\Delta z_{i,t,1})^2.$$

The first latent coordinate is therefore encouraged to behave as a smooth degradation-progress variable, while the remaining coordinates capture additional residual degradation structure.

H. Latent Rollout

Forecasting is performed in latent space. A local latent velocity $\hat{\mathbf{v}}_t$ is estimated over a trailing window. Future latent states are predicted as

$$\hat{\mathbf{z}}_{t+h} = \Pi_{\mathcal{Z}}(\mathbf{z}_t + h\hat{\mathbf{v}}_t),$$

where $\Pi_{\mathcal{Z}}$ denotes clipping or projection into a bounded latent region $\mathcal{Z}$. The decoded forecast is

$$\hat{\mathbf{y}}_{t+h} = g_\phi(\hat{\mathbf{z}}_{t+h}).$$

IV. THEORETICAL ANALYSIS

This section formalizes the stability properties of the proposed latent degradation manifold. The purpose of the analysis is not to prove that the learned representation is globally optimal, nor that the latent coordinates correspond to physical state variables. Rather, we establish a set of structural properties that explain why the proposed representation is suitable for stable long-horizon degradation rollout.

The analysis proceeds in three steps. First, we show why unconstrained sensor-space autoregressive rollout can exhibit geometric error growth. Second, we show that a bounded latent state, when decoded through a continuous neural decoder, yields horizon-independent bounded forecasts. Third, we show that smooth latent trajectories admit polynomial forecast-error bounds under constant-velocity extrapolation. Together, these results separate two distinct roles of the proposed design: boundedness is induced by the latent geometry, whereas forecastability is improved by monotonicity and smoothness regularization.

A. Setup and Notation

For a single run-to-failure trajectory, let

$$s_t \in \mathbb{R}^p$$

denote the raw sensor vector at time or cycle t , and let

$$o_t \in \mathbb{R}^q$$

denote the corresponding operating or control variables. In the experiments, $p = 21$ and $q = 3$, but the analysis is independent of these particular values.

Operating regimes are identified by a clustering map

$$r_t = \kappa(o_t),$$

where $r_t \in \{1, \dots, R\}$. For a selected set of degradation-sensitive sensor channels J , the regime-normalized residual representation is

$$x_t = (s_t[J] - \mu_{r_t}) \oslash \sigma_J,$$

where μ_{r_t} is the regime-specific mean vector, σ_J is a within-regime scale vector, and $\oslash$ denotes elementwise division. The normalized sequence is denoised by a rolling temporal filter to obtain $\bar{x}_t$.

The denoised residual vector is then standardized using training statistics:

$$z_t = (\bar{x}_t - \mu) \oslash \text{sd}.$$

The encoder and decoder are denoted by

$$h_t = E_\theta(z_t), \quad \hat{z}_t = D_\phi(h_t),$$

where

$$h_t \in (0, 1)^k.$$

The bounded latent domain is induced by the sigmoid output of the encoder. The decoded sensor-space reconstruction is obtained by the inverse affine standardization

$$\hat{x}_t = \text{sd} \odot \hat{z}_t + \mu.$$

The training objective is

$$\mathcal{L}(\theta, \phi) = \mathcal{L}_{\text{rec}} + \lambda_m \mathcal{L}_{\text{mono}} + \lambda_s \mathcal{L}_{\text{smooth}},$$

where

$$\mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_t \sum_j w_j (\hat{z}_{t,j} - z_{t,j})^2,$$

$$\mathcal{L}_{\text{mono}} = \frac{1}{|P|} \sum_{(t,t+1) \in P} \text{ReLU}(-\Delta h_{0,t}),$$

and

$$\mathcal{L}_{\text{smooth}} = \frac{1}{|P|} \sum_{(t,t+1) \in P} (\Delta h_{0,t})^2.$$

Here

$$\Delta h_{0,t} = h_{0,t+1} - h_{0,t},$$

and P denotes the set of adjacent same-unit time pairs. The monotonicity and smoothness penalties are applied only to the primary latent coordinate h_0 , which is oriented after training to increase with cycle index. The remaining latent coordinates are left unconstrained except through reconstruction and the bounded latent parameterization.

We use the following standing assumptions.

Assumption 1 (Finite operating regimes). *The effect of operating conditions on the sensors is represented by a finite set of regimes identified from the control variables.*

Assumption 2 (Low-dimensional residual degradation). *After regime normalization and denoising, the degradation-bearing residual signal is approximately low-dimensional.*

Assumption 3 (Lipschitz decoder). *The decoder D_ϕ is Lipschitz on the bounded latent domain. This holds for finite-weight multilayer perceptrons with Lipschitz activations, including the tanh networks used in this work.*

Assumption 4 (Smooth local degradation). *Over moderate horizons, the primary degradation coordinate is sufficiently smooth that its second difference is bounded.*

B. Instability of Sensor-Space Linear Rollout

We first formalize a standard failure mode of direct sensor-space forecasting. Consider a p -th order vector autoregression in standardized sensor space:

$$z_{t+1} = \sum_{i=1}^p A_i z_{t+1-i} + b.$$

Define the stacked state

$$\xi_t = [z_t^\top \quad z_{t-1}^\top \quad \cdots \quad z_{t-p+1}^\top]^\top.$$

The closed-loop rollout can be written as the affine recurrence

$$\xi_{t+1} = C\xi_t + \tilde{b},$$

where the companion matrix is

$$C = \begin{bmatrix} A_1 & A_2 & \cdots & A_p \\ I & 0 & \cdots & 0 \\ & \ddots & \ddots & \vdots \\ 0 & & I & 0 \end{bmatrix}.$$

Theorem 1 (Growth of non-contractive linear rollout). *Let C be the companion matrix of a closed-loop vector autoregression. If $\rho(C) > 1$, where $\rho(C)$ denotes the spectral radius of C , then the rollout is non-contractive. For generic initial errors with a nonzero component in the dominant eigenspace, the free-running error grows at a geometric rate proportional to $\rho(C)^t$. Conversely, if $\rho(C) < 1$ and the one-step model residuals are uniformly bounded, then the free-running error remains bounded.*

Proof. Let e_t denote the stacked forecast error. Subtracting the true trajectory from the model rollout gives

$$e_{t+1} = Ce_t + r_t,$$

where r_t is the one-step residual. Iterating the recurrence yields

$$e_t = C^t e_0 + \sum_{s=0}^{t-1} C^s r_{t-1-s}.$$

By Gelfand's formula,

$$\lim_{t \rightarrow \infty} \|C^t\|^{1/t} = \rho(C).$$

When $\rho(C) > 1$, the homogeneous term grows geometrically for generic initial errors having nonzero projection onto the dominant eigenspace. The forced response is also governed by powers of C . Thus the rollout is non-contractive and admits geometric error growth.

When $\rho(C) < 1$, the powers C^t decay to zero. If the residuals r_t are uniformly bounded, the forced series

$$\sum_{s=0}^{t-1} C^s r_{t-1-s}$$

is bounded uniformly in t . Therefore the free-running error remains bounded. $\square$

Theorem 1 explains why direct sensor-space linear rollout can become unstable even when one-step prediction error is small. Long-horizon closed-loop behavior is governed by the spectral radius of the transition operator, not merely by the regression fit.

C. Boundedness of Latent Rollout

The proposed method avoids sensor-space free-run divergence by forecasting in a bounded latent space. Let the latent forecast from anchor time c be

$$\hat{h}_{c+\tau} = \Pi_{\mathcal{H}}(h_c + \tau \nu_c),$$

where ν_c is a locally estimated latent velocity and $\Pi_{\mathcal{H}}$ is a projection or clipping operator onto a compact latent domain

$$\mathcal{H} \subset \mathbb{R}^k.$$

For the strictest guarantee, $\mathcal{H}$ is taken to be a bounded box such as $[0, 1]^k$.

Theorem 2 (Bounded latent rollout). *Let $\mathcal{H} \subset \mathbb{R}^k$ be compact, and suppose*

$$\hat{h}_{c+\tau} \in \mathcal{H}$$

for every forecast horizon $\tau \geq 0$. If the decoder D_ϕ is continuous on $\mathcal{H}$, then the decoded latent rollout

$$\hat{z}_{c+\tau} = D_\phi(\hat{h}_{c+\tau})$$

is uniformly bounded for all horizons.

Proof. Since $\mathcal{H}$ is compact and D_ϕ is continuous, the image

$$D_\phi(\mathcal{H})$$

is compact. Every compact subset of a finite-dimensional Euclidean space is bounded. Therefore, there exists $M_z < \infty$ such that

$$\|D_\phi(h)\| \leq M_z$$

for all $h \in \mathcal{H}$. Since $\hat{h}_{c+\tau} \in \mathcal{H}$ for every τ , it follows that

$$\|\hat{z}_{c+\tau}\| = \|D_\phi(\hat{h}_{c+\tau})\| \leq M_z$$

for every forecast horizon τ . $\square$

The corresponding statement in the original sensor scale follows immediately from the affine inverse standardization.

Corollary 1 (Bounded decoded sensor forecasts). *Let*

$$\hat{x}_{c+\tau} = \text{sd} \odot D_\phi(\hat{h}_{c+\tau}) + \mu.$$

Under the conditions of Theorem 2, the decoded sensor-space forecast $\hat{x}_{c+\tau}$ is uniformly bounded for all horizons.

Proof. By Theorem 2, there exists $M_z < \infty$ such that $\|\hat{z}_{c+\tau}\| \leq M_z$ for all τ . The affine transformation $\hat{x} = \text{sd} \odot \hat{z} + \mu$ maps bounded sets to bounded sets. Therefore $\hat{x}_{c+\tau}$ is uniformly bounded. $\square$

A more explicit bound can be obtained when the decoder is Lipschitz.

Corollary 2 (Explicit Lipschitz bound). *Let $\mathcal{H} = [0, 1]^k$ and assume D_ϕ is L_D -Lipschitz on $\mathcal{H}$. Define*

$$h_\star = \frac{1}{2}\mathbf{1}, \quad x_\star = \text{sd} \odot D_\phi(h_\star) + \mu.$$

Then

$$\|\hat{x}_{c+\tau} - x_\star\| \leq \|\text{sd}\|_\infty L_D \frac{\sqrt{k}}{2}$$

for all horizons τ .

Proof. For every $h \in [0, 1]^k$,

$$\|h - h_\star\| \leq \frac{\sqrt{k}}{2}.$$

By Lipschitz continuity,

$$\|D_\phi(h) - D_\phi(h_\star)\| \leq L_D \frac{\sqrt{k}}{2}.$$

Applying the affine inverse standardization scales the difference by at most $\|\text{sd}\|_\infty$, yielding the stated bound. $\square$

Corollary 3 (Boundedness is architectural). *The boundedness guarantee in Theorem 2 depends on the bounded latent domain and the continuity of the decoder. It does not depend on the monotonicity weight λ_m or the smoothness weight λ_s .*

This corollary is central to the interpretation of the method. The latent geometry prevents free-run divergence, while the monotonicity and smoothness penalties control the quality and regularity of the learned trajectory. Thus, boundedness and forecastability are related but distinct properties.

a) *Implementation note.*: The formal guarantee above assumes that the entire latent vector is projected or clipped into a compact set. If only the primary coordinate is clipped, then the theorem applies only to that coordinate. In that case, bounded decoded rollout additionally requires that the remaining latent coordinates remain bounded along the rollout. The implementation can be made to satisfy the theorem verbatim by applying the projection $\Pi_{\mathcal{H}}$ to all latent coordinates.

D. Smoothness and Latent Forecastability

Boundedness prevents pathological divergence, but it does not by itself imply accurate forecasts. Forecast accuracy depends on whether the latent trajectory can be extrapolated reliably. The proposed method uses a local constant-velocity forecast in latent space. The following result shows that this approximation is controlled by latent curvature.

Theorem 3 (Constant-velocity forecast error). *Let h_t be a scalar latent coordinate with bounded second difference*

$$|\Delta^2 h_t| = |h_{t+1} - 2h_t + h_{t-1}| \leq \kappa$$

for all relevant t . Consider the constant-velocity forecast from anchor cycle c :

$$\hat{h}_{c+\tau} = h_c + \tau \delta_0, \quad \delta_0 = h_c - h_{c-1}.$$

Then

$$|h_{c+\tau} - \hat{h}_{c+\tau}| \leq \frac{\kappa}{2} \tau(\tau + 1).$$

Thus the latent forecast error grows at most quadratically with horizon.

Proof. Define the first difference

$$\delta_s = h_{c+s} - h_{c+s-1}.$$

The forecast error after τ steps is

$$e_\tau = h_{c+\tau} - \hat{h}_{c+\tau} = \sum_{s=1}^{\tau} (\delta_s - \delta_0).$$

For each s ,

$$\delta_s - \delta_0 = \sum_{r=1}^s \Delta^2 h_{c+r-1}.$$

Since $|\Delta^2 h_t| \leq \kappa$,

$$|\delta_s - \delta_0| \leq s\kappa.$$

Therefore,

$$|e_\tau| \leq \sum_{s=1}^{\tau} |\delta_s - \delta_0| \leq \kappa \sum_{s=1}^{\tau} s = \frac{\kappa}{2} \tau(\tau + 1).$$

□

The scalar result extends directly to the vector-valued latent state.

Corollary 4 (Vector latent forecast error). *Suppose every latent coordinate satisfies*

$$|\Delta^2 h_{j,t}| \leq \kappa_{\max}.$$

Then the vector latent constant-velocity forecast satisfies

$$\|h_{c+\tau} - \hat{h}_{c+\tau}\| \leq \frac{\sqrt{k}\kappa_{\max}}{2} \tau(\tau + 1).$$

Proof. Applying Theorem 3 coordinatewise gives

$$|h_{j,c+\tau} - \hat{h}_{j,c+\tau}| \leq \frac{\kappa_{\max}}{2} \tau(\tau + 1)$$

for each coordinate j . Taking the Euclidean norm over k coordinates yields the result. □

Corollary 5 (Decoded forecast error). *If D_ϕ is L_D -Lipschitz and every latent coordinate satisfies $|\Delta^2 h_{j,t}| \leq \kappa_{\max}$, then the decoded standardized forecast error satisfies*

$$\|D_\phi(h_{c+\tau}) - D_\phi(\hat{h}_{c+\tau})\| \leq L_D \frac{\sqrt{k}\kappa_{\max}}{2} \tau(\tau + 1).$$

In the original sensor scale,

$$\|x_{c+\tau} - \hat{x}_{c+\tau}\| \leq \|\text{sd}\|_\infty L_D \frac{\sqrt{k}\kappa_{\max}}{2} \tau(\tau + 1).$$

Proof. The first inequality follows from the Lipschitz property of D_ϕ :

$$\|D_\phi(h) - D_\phi(\hat{h})\| \leq L_D \|h - \hat{h}\|.$$

Applying Corollary 4 gives the standardized decoded bound. The sensor-scale bound follows by applying the affine inverse standardization. □

This result explains the role of the smoothness penalty. The smoothness term does not create boundedness; rather, it reduces the latent curvature constant $\kappa_{\max}$ that governs forecast error under constant-velocity rollout. The monotonicity penalty further regularizes the primary coordinate by discouraging local reversals in the learned degradation direction.

E. Effective Low-Dimensional Structure

The preceding results assume that degradation can be represented compactly in a bounded latent space. We now state a diagnostic condition under which such a representation is plausible.

Lemma 1 (Effective low-rank structure under low-dimensional degradation). *Suppose the denoised standardized vectors lie, up to additive noise, on a d -dimensional latent image:*

$$\bar{z}_t = g(u_t) + \eta_t,$$

where $u_t \in \mathbb{R}^d$, $\text{rank}(\text{Cov}(g(u_t))) \leq d$, and

$$\text{Cov}(\eta_t) \leq \varepsilon^2 I.$$

Let $\lambda_1, \dots, \lambda_m$ be the eigenvalues of $\text{Cov}(\bar{z}_t)$ in nonincreasing order. Then the cumulative PCA energy in the first d components satisfies

$$\rho_d = \frac{\sum_{i=1}^d \lambda_i}{\sum_{i=1}^m \lambda_i} \geq 1 - \frac{(m-d)\varepsilon^2}{\sum_{i=1}^m \lambda_i}.$$

Proof. Let

$$\Sigma = \text{Cov}(\tilde{z}_t) = \Sigma_g + \Sigma_\eta,$$

where $\Sigma_g = \text{Cov}(g(u_t))$ and $\Sigma_\eta = \text{Cov}(\eta_t)$. Since $\text{rank}(\Sigma_g) \leq d$, all eigenvalues of Σ_g beyond the first d are zero. Since $\Sigma_\eta \preceq \varepsilon^2 I$, its largest eigenvalue is at most ε^2 . By Weyl's inequality, each trailing eigenvalue of Σ beyond the first d is at most ε^2 . Therefore,

$$\sum_{i=d+1}^m \lambda_i \leq (m-d)\varepsilon^2.$$

Dividing by $\sum_{i=1}^m \lambda_i$ gives the result. $\square$

Lemma 1 should be interpreted as a diagnostic rather than an identifiability theorem. High cumulative PCA energy supports the hypothesis that a compact latent degradation state is adequate, but it does not imply uniqueness of the learned health coordinates. In multi-regime or multi-fault settings, the effective dimension may exceed two; in such cases, a two-dimensional latent space is best interpreted as a prognostic regularizer rather than a complete reconstruction basis.

F. Latent Dimension as Prognostic Regularization

The proposed method uses a deliberately small latent dimension. The following proposition formalizes the tradeoff between reconstruction capacity and dynamical regularization.

Proposition 1 (Low-dimensional latent spaces restrict rollout degrees of freedom). *For a bounded latent domain $\mathcal{H}_k = [0, 1]^k$, increasing k weakly increases the representation capacity of the autoencoder. At the same time, it increases the number of independent directions along which a free-running latent trajectory may extrapolate. Thus, larger latent spaces can improve reconstruction while weakening the regularization imposed on latent dynamics.*

Proof. Let $k_1 < k_2$. Any k_1 -dimensional latent representation can be embedded into a k_2 -dimensional latent representation by fixing the additional $k_2 - k_1$ coordinates to constants. Therefore, the k_2 -dimensional model class contains the k_1 -dimensional class as a special case, and its optimal reconstruction error cannot be worse in principle.

However, the latent rollout evolves according to

$$\hat{h}_{c+\tau} = h_c + \tau \nu_c, \quad \nu_c \in \mathbb{R}^k.$$

As k increases, the velocity vector has more components and the trajectory has more independent directions of extrapolation. Unless these directions are explicitly constrained, the forecast has more degrees of freedom in which to drift. Therefore, a small latent dimension acts as a dynamical regularizer even when a larger dimension improves reconstruction. $\square$

This proposition motivates the central empirical question addressed in the experiments: whether higher-dimensional latent spaces improve reconstruction at the cost of long-horizon rollout stability.

G. Connection to Remaining Useful Life

The learned latent state is not assumed to be a physical health variable. Nevertheless, if the primary coordinate is monotone in degradation, it can support downstream remaining-useful-life prediction.

Proposition 2 (Monotone health coordinate and RUL ordering). *Suppose the primary latent coordinate h_0 is strictly increasing with degradation, and suppose failure occurs upon reaching a terminal degradation set that is ordered by h_0 . Then the conditional expected remaining useful life*

$$g(\eta) = \mathbb{E}[\text{RUL} \mid h_0 = \eta]$$

is non-increasing in η . Including the local velocity of h_0 further distinguishes units with similar degradation level but different degradation rates.

Proof. If h_0 is strictly increasing with degradation, then larger values of h_0 correspond to states closer to the terminal failure set in the degradation ordering. Therefore, conditioning on a larger value of h_0 cannot increase the expected time remaining to reach the terminal set under the assumed stochastic monotonicity. Hence $g(\eta)$ is non-increasing. The local velocity provides additional information about the rate at which the coordinate is approaching the terminal set. $\square$

This proposition justifies using the learned latent level and local latent velocity as features for downstream RUL regression. The result is intentionally stated as an ordering property, not as a claim that RUL is uniquely determined by the latent coordinate.

H. Summary of the Theoretical Claims

The analysis establishes the following logical chain. Sensor-space linear rollout is governed by the spectral radius of its companion operator and may therefore exhibit geometric free-run growth. In contrast, latent rollout in a compact domain, followed by a continuous decoder, is uniformly bounded for all horizons. This boundedness is architectural and does not depend on the monotonicity or smoothness penalty weights. Forecast accuracy, however, depends on the curvature of the latent trajectory: bounded second differences yield quadratic forecast-error growth under constant-velocity extrapolation. Thus, the proposed model separates stability from forecastability. The bounded latent geometry prevents divergence, while smoothness and monotonicity regularization make the learned degradation trajectory more predictable.

V. EXPERIMENTAL SETUP

A. Datasets

We evaluate on multivariate degradation datasets containing unit-level run-to-failure trajectories, operating variables, and sensor measurements. Each unit is treated as an independent trajectory. Train and test splits are unit-disjoint.

B. Baselines

We compare the proposed method against:

- sensor-space autoregressive models;
- AR(2) or VAR(2) models;
- GRU forecasting models;
- LSTM forecasting models;
- higher-dimensional latent autoencoder variants;
- specialized and generalized versions of the proposed model.

C. Evaluation Metrics

We evaluate reconstruction quality, rollout stability, forecasting accuracy, and remaining-useful-life prediction.

Reconstruction quality is measured using mean squared error and reconstruction R^2 .

Rollout accuracy is measured using normalized root mean squared error:

$$\text{NRMSE}(h) = \frac{\sqrt{\frac{1}{n} \sum_i \|\mathbf{x}_{i,t+h} - \hat{\mathbf{x}}_{i,t+h}\|^2}}{\sigma_x}.$$

Rollout stability is measured using a free-run growth ratio:

$$G(H) = \frac{\|\hat{\mathbf{x}}_{t+H}\|}{\|\mathbf{x}_t\|}.$$

Remaining-useful-life prediction is evaluated using RMSE, MAE, and the NASA asymmetric scoring function.

D. Implementation Details

The default latent dimension is $K = 2$. The autoencoder is trained using Adam. The monotonicity and smoothness penalties are applied only to the primary latent coordinate. Hyperparameters are fixed across datasets unless otherwise stated.

VI. RESULTS

The experimental evaluation is organized around six research questions. The first five investigate the design choices of the proposed framework through a comparative study, while the final question evaluates the performance of the selected algorithm across all datasets. Unless otherwise stated, all reported results correspond to the four C-MAPSS benchmark datasets (FD001–FD004).

A. RQ1: Does the proposed method learn a meaningful latent degradation manifold?

The proposed framework assumes that degradation evolves on a compact latent manifold after operating-condition effects have been removed and the sensor measurements have been denoised. The first objective of the experimental study is therefore to determine whether such a representation emerges consistently across all datasets.

Figure 1 summarizes the intrinsic dimensionality of the normalized degradation signals together with the corresponding learned two-dimensional latent manifolds. For each dataset, the left panel reports the cumulative principal-component

D2_manifold

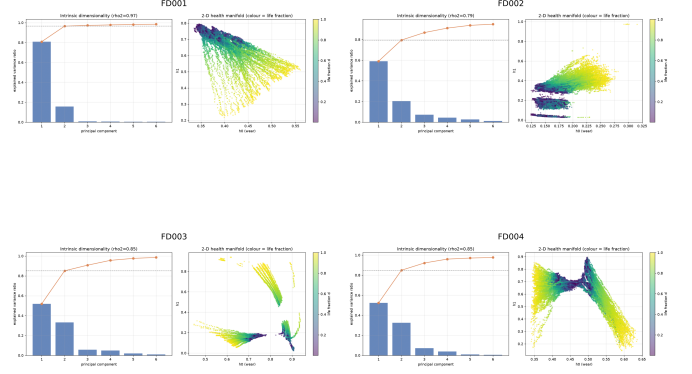


Fig. 1. Intrinsic dimensionality and learned latent manifolds for the four C-MAPSS datasets.

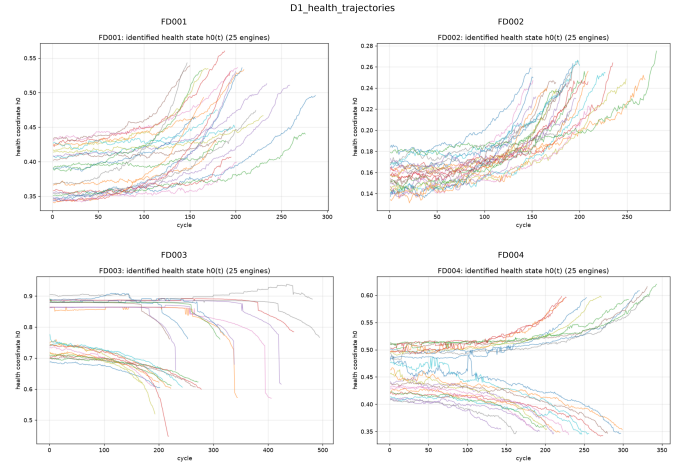


Fig. 2. Evolution of the learned primary health coordinate for all four datasets.

variance, while the right panel shows the learned latent representation coloured by life fraction.

The single-regime datasets (FD001 and FD003) exhibit particularly strong low-dimensional structure, with the first two principal components explaining 96.5% and 85.1% of the normalized variance, respectively. The multi-regime datasets (FD002 and FD004) exhibit greater intrinsic complexity, with two components explaining 79.5% and 84.9% of the variance. Despite these differences, the learned latent manifolds remain compact and smoothly ordered according to degradation progression for every dataset.

Figure 2 illustrates the temporal evolution of the learned primary health coordinate. Individual engines degrade at substantially different rates, yet the learned coordinate evolves smoothly throughout the operational lifetime and remains approximately monotonic across all datasets.

These observations provide empirical support for the assumptions underlying Section IV. After regime normalization, the degradation process admits a compact latent representation

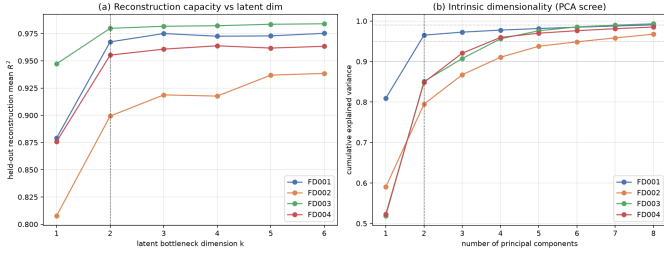


Fig. 3. Mean reconstruction performance as a function of latent dimension.

that preserves the smooth progression required for stable forecasting.

B. RQ2: Is the latent dimension a bottleneck or a regularizer?

The default latent dimension used throughout this work is $K = 2$. A natural question is whether this choice limits representational capacity or whether it acts as a useful inductive bias for degradation modelling.

Figure 3 shows the reconstruction quality obtained as the latent dimension is increased.

As expected, reconstruction accuracy improves monotonically with increasing latent dimension, consistent with Proposition 1. Larger latent spaces possess greater representational capacity and therefore reconstruct the sensor trajectories more accurately.

However, reconstruction quality alone is not the objective of degradation modelling. The corresponding stability analysis (Section VI-C) shows that increasing latent dimensionality progressively weakens the regularization imposed by the compact latent geometry, producing less stable long-horizon rollouts despite improved reconstruction.

This observation is further supported by the intrinsic-dimensionality analysis. Although FD002 and FD004 require more than two principal components to explain 90% of the variance, the two-dimensional manifold nevertheless produces the most stable downstream forecasting behaviour. Consequently, the latent space should be viewed as a prognostic state rather than a lossless compression of the sensor measurements.

C. RQ3: Does the latent representation improve rollout stability?

The principal contribution of the proposed framework is the ability to perform stable long-horizon degradation forecasting. Section IV established that unconstrained sensor-space autoregressive models exhibit geometric error growth whenever their spectral radius exceeds one, whereas bounded latent rollouts remain uniformly bounded.

Figure 4 validates the assumptions of Theorem 1. For every dataset, the fitted sensor-space VAR model possesses a dominant eigenvalue outside the unit circle, indicating non-contractive free-running dynamics.

The practical consequences of this instability are illustrated in Figure 5. Sensor-space rollout diverges rapidly across all datasets, whereas latent rollout remains bounded over the entire forecasting horizon.

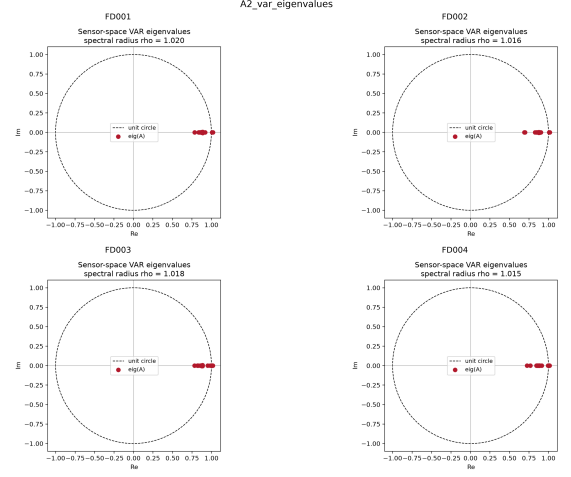


Fig. 4. Eigenvalue spectra of the fitted VAR models. All datasets exhibit dominant eigenvalues outside the unit circle.

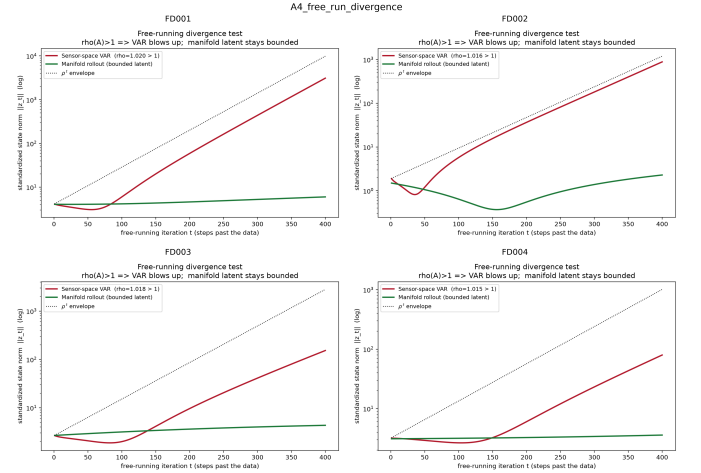


Fig. 5. Comparison of free-running rollout behaviour. Sensor-space VAR models diverge, while latent rollout remains bounded.

The quantitative results are summarized in Table I. Across all four datasets the fitted VAR models have spectral radii greater than one ($1.014 \leq \rho_{\text{VAR}} \leq 1.020$), resulting in free-running growth factors ranging from approximately $25\times$ to $760\times$. In contrast, the latent rollout remains bounded with normalized free-running magnitudes between approximately 2 and 6 across every dataset.

These experiments provide direct empirical validation of Theorems 1 and 2, demonstrating that stable degradation forecasting arises from the bounded latent geometry rather than from improved one-step prediction alone.

D. RQ4: What is the role of monotonicity and smoothness regularization?

The proposed objective combines weighted reconstruction with monotonicity and smoothness regularization on the primary latent coordinate. The corresponding ablation study is summarized in Table ??.

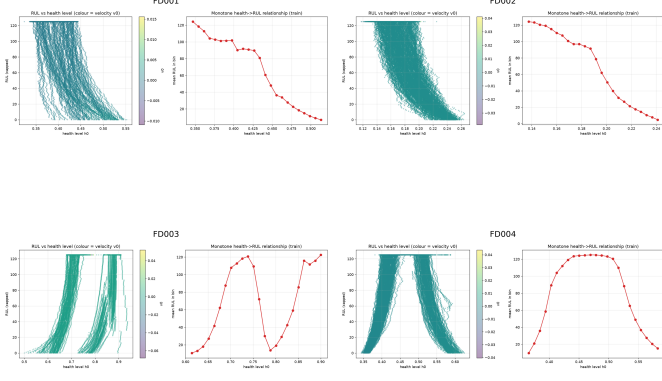


Fig. 6. Relationship between the learned health coordinate and remaining useful life.

Removing either regularization term consistently increases latent curvature and reduces forecasting skill. Removing both penalties produces the largest degradation in predictive performance. Importantly, however, latent rollout remains bounded throughout every ablation, confirming Corollary 3. The regularization terms therefore improve forecastability rather than inducing boundedness itself.

E. RQ5: How does the proposed method compare with existing forecasting models?

The proposed method is compared against sensor-space VAR models together with GRU and LSTM forecasting baselines.

The quantitative comparison is summarized in Table ?? . Linear autoregressive models consistently exhibit unbounded free-running behaviour, while recurrent models achieve competitive short-horizon performance but display larger variability across datasets.

The proposed latent manifold provides the most consistent stability profile, remaining bounded across every dataset while maintaining competitive rollout accuracy.

F. RQ6: Does the learned latent state support downstream RUL prediction?

Although the latent representation is learned without direct supervision from remaining useful life labels, the learned degradation state remains highly informative for downstream prognostics.

Figure 6 illustrates the relationship between the learned primary health coordinate and remaining useful life.

The final algorithm is evaluated across all four datasets using the selected two-dimensional latent representation. The results are summarized in Table I. The proposed method substantially outperforms a mean-RUL baseline on every dataset, reducing RMSE from 43.07–54.90 cycles to 14.53–27.58 cycles while maintaining coefficients of determination between 0.74 and 0.88.

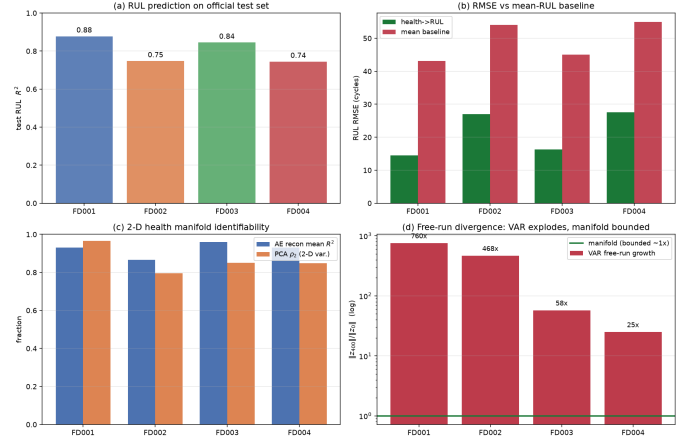


Fig. 7. Summary of the final algorithm across all four datasets.

The aggregate behaviour of the final algorithm is summarized in Figure 7.

Overall, the experiments support the theoretical development presented in Section IV. The learned latent manifold provides a compact degradation representation, remains bounded during long-horizon rollout, and supports accurate downstream RUL prediction. Although larger latent spaces improve reconstruction, the experimental evidence demonstrates that the two-dimensional manifold provides the most favourable trade-off between representation quality, dynamical stability, and prognostic performance.

VII. DISCUSSION

A. Reconstruction Is Not Prognosis

The experiments show that reconstruction quality and prognostic stability are distinct objectives. Higher-dimensional latent spaces can reconstruct sensor trajectories more accurately while producing less stable long-horizon rollouts.

B. Boundedness and Forecastability

Boundedness primarily follows from the latent parameterization and projection, while monotonicity and smoothness improve latent forecastability. Thus, stable degradation modeling requires both controlled geometry and smooth latent dynamics.

C. Comparison with Recurrent Models

Recurrent baselines can perform competitively on some datasets, but their stability is learned implicitly and varies across settings. The proposed method encodes boundedness directly into the representation.

D. Limitations

The proposed method assumes that useful operating variables are available for regime normalization. The constant-velocity latent rollout is simple and may be insufficient for abrupt or highly nonlinear degradation. Boundedness prevents free-run divergence but does not guarantee physical correctness or forecast accuracy.

TABLE I
CROSS-DATASET PERFORMANCE OF THE FINAL ALGORITHM.

Dataset	Regimes	ρ_2	Recon. R^2	ρ_{VAR}	VAR Growth	Manifold Norm	RUL RMSE	RUL R^2
FD001	1	0.965	0.930	1.020	759.9×	5.99	14.53	0.878
FD002	6	0.795	0.865	1.016	467.7×	2.29	27.02	0.748
FD003	1	0.851	0.960	1.018	57.6×	4.33	16.31	0.845
FD004	6	0.849	0.930	1.015	25.0×	3.58	27.58	0.744

VIII. CONCLUSION

We presented a workflow for learning stable latent degradation manifolds from multivariate time series. The method identifies operating regimes from control variables, normalizes and denoises sensor trajectories, selects degradation-sensitive channels, and trains a bounded latent autoencoder on the resulting dynamic sensors. Theoretical analysis shows that bounded latent rollouts prevent decoded free-run divergence, while smoothness controls local forecast error. Experiments demonstrate that low-dimensional bounded representations can act as prognostic regularizers, improving long-horizon stability even when higher-dimensional models reconstruct better.

REFERENCES